

Every Threshold Has a Place: Conserved Spatial Evidence Ledgers for Auditable Ordinal Vision Supplementary Material

Anonymous CVPR submission

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1. Scope of this supplement

This supplement makes the completed protocol, numerical, grading, and retrospective mechanism evidence inspectable in one place. It also documents the exact contracts of the two declared ordinal decoders and the corrective IDRiD protocol. It does not fill unfinished experiments with pilot values. At the time of this draft, the replicated intrinsic-baseline matrix, the shortcut audit, the validation-wide grouped-intervention gate, and the corrective IDRiD-v2 deletion statistics remain under checksum-sealed release gates. Their table slots are marked pending; no conclusion in the main paper depends on assuming a favorable outcome.

The main paper supplies the architecture and notation. Briefly, the only image-dependent input to the ordinal decoder is a nonnegative spatial ledger ℓ_{nsuk} indexed by image n , scale s , cell u , and adjacent ordinal boundary k . The total boundary rate is

$$\lambda_{nk} = \pi_k + \sum_s \sum_u \ell_{nsuk}, \quad \ell_{nsuk} \geq 0, \quad (1)$$

where π_k is image independent. Group deletion subtracts selected stored entries and replays the same deterministic decoder. Exact replay therefore belongs to the conserved ledger, not uniquely to a matrix exponential.

2. Decoder contracts and numerical verification

2.1. Primary pure-birth CTMC decoder

For $K = 5$ grades, $B = K - 1 = 4$ rates define the upper-bidiagonal generator

$$Q_{k,k} = -\lambda_k, \quad Q_{k,k+1} = \lambda_k, \quad Q_{K-1,K-1} = 0. \quad (2)$$

At exposure t , its posterior is

$$\mathbf{p}_{\text{ctmc}}(t) = e_0^\top \exp(tQ). \quad (3)$$

The deployed model fixes $t = 1$. Rates are bounded by construction to the audited domain $[0, 64]^4$; neither the model nor our expressiveness statement uses unbounded rates. The implementation evaluates a degree-24 Taylor polynomial after scaling Q by 2^{-s} , where

$$s = \max \left\{ 0, \left\lceil \log_2 \left(\frac{\max_n \|Q_n\|_\infty}{0.5} \right) \right\rceil \right\}, \quad (4)$$

and then squares s times in FP64. There is no posterior clipping or renormalization.

2.2. Same-ledger sequential-hazard decoder

The matched control consumes the identical four rates. Its adjacent hazards, cumulative tails, and normalized posterior are

$$h_k(t) = 1 - \exp(-t\lambda_k), \quad (5)$$

$$T_k(t) \equiv P_{\text{seq}}(Y > k) = \prod_{j=0}^k h_j(t), \quad (6)$$

$$p_y^{\text{seq}}(t) = \begin{cases} (1 - h_y(t)) \prod_{j < y} h_j(t), & y < K - 1, \\ \prod_{j=0}^{K-2} h_j(t), & y = K - 1. \end{cases} \quad (7)$$

The class probabilities telescope to one. Both declared decoders are coordinatewise monotone in the cumulative tails and inherit exact frozen-ledger deletion and retention. Only the CTMC defines one homogeneous shared-clock transition kernel satisfying $P(t + r) = P(t)P(r)$. The natural restarted sequential construction generally does not obey that semigroup identity. We claim this as a mathematical contract, not as validated longitudinal disease dynamics or an endpoint accuracy advantage.

2.3. Independent numerical audits

The deployed FP64 CTMC routine was compared against an independent 90-decimal-digit mpmath.expm reference on 138 seeded random and adversarial rate

Table 1. Attribution of guarantees. “Shared” means the property holds for both declared decoders when they consume the same conserved ledger.

Property	Ledger	CTMC	Sequential
Nonnegative local entries	✓	–	–
Exclusive image path	✓	consumes	consumes
Exact grouped replay	✓	shared	shared
Normalized posterior	–	✓	✓
Monotone ordinal tails	–	✓	✓
Homogeneous group	–	✓	no, generally

Table 2. Completed numerical reference audit.

Quantity	Maximum error
Absolute posterior error	1.4877×10^{-14}
Relative posterior error	9.4059×10^{-14}
Posterior row-sum error	1.6209×10^{-14}
Absolute gradient error	1.3323×10^{-15}
Relative gradient error	1.9054×10^{-8}

vectors spanning $[0, 64]^4$. Twelve additional cases compared autograd with independent finite differences. Table 2 reports the completed maxima. The relative gradient statistic is dominated by near-zero reference derivatives, so the absolute error is the more informative quantity. These tests validate the declared bounded implementation; they do not establish universal saturation of the categorical simplex.

A separate decoder-contract audit sampled bounded rates and interior posterior targets. The largest tested endpoint reconstruction error was 2.08×10^{-12} for the CTMC and 1.95×10^{-16} for the sequential decoder; observed monotonicity error was zero. The CTMC semigroup error was 4.55×10^{-15} , whereas the smallest recorded violation for the corresponding restarted sequential construction was 0.01119. These sampled inversion results provide no evidence of a fixed-horizon expressiveness advantage for the CTMC.

3. Freeze chronology and evaluation discipline

Table 3 separates architecture development, locked evaluation, and retrospective analysis. The bounded V3 architecture was frozen before the complete cross-validation run. Architecture and hyperparameter search used fixed inner-development splits, however, and was not independently nested inside each outer fold. Consequently, the full-CV results estimate one frozen configuration; they are not evidence from nested

model-class selection.

The exact full-CV run tag is origin_v3_full_cv_20260922T084547Z.

Within each outer training pool, 10% is reserved for inner validation. Optimization and checkpoint selection use only the training and inner splits. The rule is lexicographic: higher inner-validation accuracy, then higher QWK, then lower loss. The selected checkpoint is evaluated once on the locked outer fold; a deterministic second pass exports its posterior for auditing. EyePACS folds are grouped by patient identity parsed from paired-eye filenames. APTOS supplies no patient identifier, so its folds are image stratified. A failed EyePACS fold-9 attempt was archived and rerun from scratch under the unchanged commit, split, seed, and configuration.

4. Complete outer-fold grading results

Tables 4 and 5 give every outer fold. Accuracy and balanced accuracy are percentages; MAE uses the discrete MAP grade. ECE is 15-bin top-label expected calibration error. Pooled OOF metrics are computed once from the union of per-image exports and therefore need not equal the unweighted mean of fold values.

4.1. Pooled confusion matrices and long-tail behavior

The EyePACS pooled confusion matrix (rows are true grade, columns predicted grade) is

$$\begin{bmatrix} 25013 & 323 & 447 & 3 & 24 \\ 1335 & 697 & 410 & 0 & 1 \\ 974 & 312 & 3794 & 170 & 42 \\ 34 & 8 & 500 & 308 & 23 \\ 57 & 3 & 177 & 79 & 392 \end{bmatrix}.$$

It yields per-grade recall [96.91, 28.53, 71.69, 35.28, 55.37]%. The corresponding APTOS matrix is

$$\begin{bmatrix} 1783 & 19 & 3 & 0 & 0 \\ 30 & 186 & 145 & 0 & 9 \\ 3 & 37 & 918 & 2 & 39 \\ 0 & 0 & 136 & 14 & 43 \\ 0 & 10 & 72 & 6 & 207 \end{bmatrix},$$

with recall [98.78, 50.27, 91.89, 7.25, 70.17]%. These tables expose the minority-grade failures hidden by overall accuracy: EyePACS grade 1 and grade 3, and especially APTOS grade 3, remain poorly recognized.

5. Complete retrospective fold-9 mechanism study

This study was specified only after EyePACS fold 9 and the complete OOF aggregate had been observed.

Table 3. Development and freeze chronology. Commit IDs and signed run manifests, rather than prose recollection, are the primary provenance.

Date	Commit/run	Event	Status and visible evaluation
Sep. 8, 2026	49c541f...	Initial conserved ordinal generator	Inner-validation development runs
Sep. 9–10	b0f4609...	Bounded V3 parameterization, FP64 correction, and serializable certificate audit frozen	Final V3 architecture; no full-CV aggregate existed
Sep. 22	a9e9f65...	Immutable split, selection, and full-CV launcher frozen	Outer folds remained locked during training and selection
Sep. 22–25	full-CV run	5 APTOS and 10 patient-grouped EyePACS outer folds	Complete OOF evaluation of frozen V3
Sep. 26	290e0a0...	Ten-arm fold-9 mechanism protocol added	Retrospective: fold 9 and full aggregate had been observed
Oct. 1 onward	acceptance-revision protocols	Numerical, artifact, IDRiD, shortcut, grouped-replay, and matched-baseline audits	May test claims; may not modify V3 and reuse its OOF results as prospective evidence

Table 4. Complete patient-grouped EyePACS 10-fold outer-CV results.

Fold	N	Acc.	MAP-MAE	QWK	Bal. acc.	Macro-F1	ECE
0	3518	85.759	0.1961	0.8242	56.774	0.6098	0.0384
1	3518	86.356	0.1873	0.8295	56.428	0.6135	0.0273
2	3514	85.714	0.2046	0.8027	57.969	0.6251	0.0318
3	3514	85.544	0.2103	0.7857	53.982	0.5950	0.0248
4	3514	86.027	0.2026	0.8030	56.519	0.6220	0.0333
5	3510	86.752	0.1744	0.8510	60.305	0.6470	0.0405
6	3510	86.382	0.1869	0.8249	57.876	0.6342	0.0265
7	3510	86.154	0.1923	0.8203	61.016	0.6467	0.0372
8	3510	85.926	0.1991	0.8149	59.198	0.6290	0.0281
9	3508	85.262	0.2092	0.7961	55.286	0.5893	0.0218
Fold mean	—	85.988	0.1963	0.8152	57.535	0.6212	0.0310
Sample SD	—	0.442	0.0113	0.0189	2.195	0.0196	0.0063
Pooled OOF	35126	85.988	0.1963	0.8153	57.557	0.6223	0.0288

It therefore provides retrospective mechanism evidence, not a confirmatory benchmark. All ten variants share the same patient split, ImageNet initialization policy, optimizer, augmentation, checkpoint rule, and one training-stream seed. The locked outer fold contains 3,508 images from 1,754 paired-eye patient clusters. Confidence intervals use 10,000 paired patient-cluster bootstrap resamples; no multiplicity correction is applied.

The pooled continuation-ratio model has the lowest MAP-MAE and its paired interval excludes zero, while full Origin has better ECE by 0.0273 ([0.0198, 0.0357] when expressed as pooled-minus-Origin). The fine-only variant is decisively worse, whereas the coarse-only interval spans zero. The same-ledger decoder comparison does not show a CTMC grading advantage. These results motivate, but cannot substitute for, the replicated

matched baseline suite listed in Sec. 8.

6. IDRiD semantic-association protocol

6.1. Initial v1 alignment analysis; corrective v2 pending

The external analysis uses all 81 images in the IDRiD segmentation subset [1]: 54 training-split and 27 testing-split images. All ten independently inner-selected EyePACS outer-fold checkpoints are evaluated; no IDRiD image, mask, or statistic is used for training, checkpoint selection, hyperparameter tuning, or scale selection.

For each image we form the union of the official microaneurysm, hemorrhage, hard-exudate, and soft-exudate masks. The dominant-field-of-view crop is computed from the image and applied identically to

Table 5. Complete image-stratified APTOS 5-fold outer-CV results.

Fold	N	Acc.	MAP-MAE	QWK	Bal. acc.	Macro-F1	ECE
0	733	84.584	0.2019	0.9051	61.905	0.6103	0.0838
1	733	84.175	0.2005	0.9046	60.171	0.6077	0.0563
2	733	84.993	0.1855	0.9218	67.468	0.6903	0.0443
3	732	85.109	0.1899	0.9155	66.056	0.6499	0.0315
4	731	85.499	0.1902	0.9115	62.749	0.6405	0.0777
Fold mean	—	84.872	0.1936	0.9117	63.670	0.6397	0.0587
Sample SD	—	0.508	0.0072	0.0073	3.013	0.0337	0.0220
Pooled OOF	3662	84.872	0.1936	0.9118	63.673	0.6438	0.0502

Table 6. Complete fold-9 mechanism results. Deltas and 95% intervals are candidate minus full Origin. Lower MAP-MAE and ECE are better; higher remaining metrics are better.

Variant	Params (M)	Acc.	Δ Acc. [95% CI]	MAP-MAE	Δ MAE [95% CI]	QWK	Bal. acc.	ECE
Full Origin	28.01	85.66	reference	0.2007	reference	0.8132	58.22	0.0429
Pooled softmax	28.07	85.86	+0.20[−0.57, 0.97]	0.1958	−0.0048[−0.0177, 0.0080]	0.8196	59.83	0.0597
Pooled cumulative	28.07	85.72	+0.06[−0.74, 0.86]	0.1978	−0.0029[−0.0151, 0.0094]	0.8210	57.10	0.0459
Same-ledger hazard	28.01	85.86	+0.20[−0.48, 0.88]	0.1984	−0.0023[−0.0145, 0.0100]	0.8120	57.26	0.0357
Direct simplex atoms	28.01	86.00	+0.34[−0.40, 1.05]	0.1984	−0.0023[−0.0145, 0.0103]	0.8110	59.53	0.0430
NLL only	28.01	86.23	+0.57[−0.06, 1.20]	0.1956	−0.0051[−0.0160, 0.0057]	0.8061	58.11	0.0441
Fine scales (s_4, s_8)	27.86	83.69	−1.97[−2.88, −1.08]	0.2377	+0.0371[0.0217, 0.0530]	0.7544	42.10	0.0132
Coarse scales (s_{16}, s_{32})	27.97	85.66	+0.00[−0.66, 0.68]	0.2052	+0.0046[−0.0071, 0.0162]	0.7987	57.48	0.0493
Independent sigmoid atoms	28.01	85.80	+0.14[−0.54, 0.83]	0.2013	+0.0006[−0.0108, 0.0123]	0.8094	59.22	0.0437
Pooled continuation ratio	28.07	86.40	+0.74[−0.06, 1.54]	0.1861	−0.0145[−0.0274, −0.0017]	0.8299	62.48	0.0702

image and masks. Images are resized bilinearly; masks use nearest-neighbor interpolation. For each native feature lattice, adaptive max pooling marks a bin positive iff any aligned lesion-mask pixel lies in its spatial support. At true grade y , the scalar score for each cell is its stored local ledger rate at

$$b(y) = \min\{\max(y - 1, 0), 3\}. \quad (8)$$

Average precision (AP) is computed separately for every image $\times$ checkpoint $\times$ scale. Checkpoints and scales are then averaged within image, and image is the inferential cluster. This order makes the test within-image: it cannot obtain positive enrichment merely because one image has both more lesions and globally larger evidence.

V1 reported fine-scale AP 0.1861 (95% image-cluster CI [0.1637, 0.2095]) at prevalence 0.0815 and within-image AP-minus-prevalence 0.1045 ([0.0938, 0.1159]). A later code audit found that its element-wise stable sort did not group exact score ties at common thresholds, so these values are provisional. V2 recomputes non-interpolated threshold AP with exact ties grouped; no AP claim is promoted until that terminal manifest exists.

6.2. Corrective bidirectional deletion protocol (v2; pending)

The completed v1 fine-scale replay result used exact within-image/scale count matching and found a guided-minus-control increase of 0.0111 in $|\Delta E[Y]|$ ([0.0097, 0.0125]). A later census found that v1 could not match all lesion-positive cells at 14 dense image $\times$ coarse-scale cases: there were fewer nonlesion cells than lesion cells. The fine deletion result remains exactly matched, but v1 coarse/all-scale deletion lifts are demoted to exploratory; the separate AP issue above is also corrected by v2.

The checksum-sealed v2 reanalysis corrects this without retraining. For image i and scale s , let L_{is} and C_{is} denote unique lesion-positive and nonlesion cells. Each of 20 deterministic paired repeats uses

$$m_{is} = \min(|L_{is}|, |C_{is}|) \quad (9)$$

and samples m_{is} unique cells from each set. Guided and control groups therefore have identical per-scale cell counts, scale composition, geometry exposure R/N_s , nominal stride-area cost, and target-boundary pre-atom multiplier. The actual stored rate mass removed is intentionally not matched: it is the outcome being tested. Retinal anatomy and eccentricity are not matched. Aggregation is repeats within checkpoint and image, checkpoints within image, then image-cluster inference. The primary contrast is all-scale guided minus

bidirectionally matched control deletion; fine/coarse, grade, boundary, segmentation-split, and lesion-type strata are secondary and descriptive. Protocol checksum:

d99ec88a9c83533d825c954d49061d88
d349a14a23d55fe15105c2afee74e414

The v2 effect estimate and interval remain pending until sealed release. Until its terminal manifest exists, this document makes no positive claim from coarse/all-scale deletion.

7. Artifact and release manifest

The completed base manifest indexes the 15 selected V3 checkpoints and 15 compressed fold-membership artifacts and binds them to source, architecture, configuration, implementation, split, and SHA-256 signatures. Its manifest checksum is

fa56cd1d25a35dedf09b7f65a86f1c5
91ef0ab9d434310256cd8dd1fa4d6e59d

Hashes establish identity only when the corresponding artifact is supplied; they do not by themselves enable re-inference.

The anonymous package contract therefore requires the following concrete contents, without redistributing licensed fundus pixels:

1. source commit, environment, architecture/configuration signatures, and selected checkpoint tensors where hosting permits, otherwise stable archival links plus SHA-256;
2. deterministic split code and compressed memberships containing hashed image identity, hashed patient cluster where available, label, and role;
3. every OOF label, posterior, cumulative tail, MAP decision, expected grade, fold, and hashed cluster;
4. per-image/per-boundary intervention curves, group masks or deterministic control seeds, protocol hashes, and audit scripts;
5. numerical-reference and decoder-contract inputs and outputs;
6. IDRiD records and, if they pass their gates, shortcut and matched intrinsic-baseline records; and
7. one top-level fail-closed manifest linking every reported table to a checksummed aggregate.

Any missing fold, split overlap, duplicate outer image, mismatched hash, incomplete randomized control, or nonfinite value invalidates the affected aggregate. The final assembled anonymous archive is pending dependency-gated completion of the experiments in [Sec. 8](#); the present document does not imply that this archive has already been released publicly.

Table 7. Reserved experiments. Every result is intentionally withheld until its complete release gate passes.

Experiment	Predeclared question and status
Matched base-lines	Does the ledger improve a concrete audit over same-ledger hazards, pooled continuation ratio, ordinal Additive-MIL, and sparse local-logit analogues while retaining grading? Pending sealed release; no result claimed.
Grouped re-play	Do ranked groups show necessity and sufficiency beyond scale-, footprint-, count-, and mass-matched controls? Pending sealed release; no result claimed.
Ordinal short-cut	Does the ledger identify a known cue’s location and boundary, and does internal deletion agree with pixel toggles? Pending sealed release; no result claimed.
IDRiD v2	Does tie-grouped AP remain above within-image prevalence, and does all-scale guided deletion exceed within-image/scale matched controls after correcting 14 dense cases? Pending sealed release; protocol fixed in Sec. 6 .
Artifact assembly	Can promoted results be regenerated from predictions, memberships, records, scripts, and hashes? Pending sealed release on upstream terminal artifacts.

8. Reserved acceptance-revision results

[Table 7](#) records the predeclared questions without converting running jobs into evidence. The paper will be updated only from terminal, checksummed aggregate manifests. If a gate fails, the corresponding claim will be withdrawn rather than tuned post hoc on the same outer predictions.

9. Interpretation checklist

The following language is supported by completed evidence:

- the ledger is the exclusive image-dependent prediction path;
- ledger entries are nonnegative, boundary indexed, and exactly conserved;
- grouped frozen-ledger deletion/retention is exact for either declared deterministic decoder;
- the deployed CTMC is numerically accurate on its bounded rate domain;
- complete CV supports competitive overall grading under the stated protocols.

The following language is not supported by the completed evidence and is not used: pixel-causal explanation, automatic lesion recognition, clinical necessity, a demonstrated CTMC endpoint advantage, solved class imbalance, fully nested architecture selection, finalized IDRiD association, or superiority to intrinsic local predictors. Those boundaries are part of the method’s scientific contract, not presentation caveats.

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References

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- [1] Prasanna Porwal, Samiksha Pachade, Manesh Kokare, Girish Deshmukh, Jaemin Son, Woong Bae, Liangchen Liu, Jianzong Wang, Xinhui Liu, Lie Gao, TianBao Wu, Jing Xiao, Fengyan Wang, Baocai Yin, Yi Wang, Gopichandh Danala, Lili He, Yoon Ho Choi, Yong Jun Lee, Sang-Hyuk Jung, Zhiyuan Li, Xiaoke Sui, Junyan Wu, Xiaolong Li, Ting Zhou, Janos Toth, Andras Hajdu, Mathieu Lamard, Gwenole Quellec, and Fabrice Meriaudeau. IDRiD: Diabetic retinopathy—segmentation and grading challenge. *Medical Image Analysis*, 59: 101561, 2020. 3